Haze Removal And Ship Tracking In Hazy Weather Based On Deep Learning

Zhenzhen Zhou
College of Merchant Marine
Shanghai Maritime University
Shanghai, China
zhouzhenzzzmail@163.com

Shiqi Geng
College of Merchant Marine
Shanghai Maritime University
Shanghai, China
garvis_gsq@163.com

Jiansen Zhao*
College of Merchant Marine
Shanghai Maritime University
Shanghai, China
jszhao@shmtu.edu.cn

2023 7th International Conference on Transportation Information and Safety (ICTIS) | 979-8-3503-0853-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICTIS60134.2023.10243692

***Abstract*—When the tracking algorithm is applied to a hazy maritime image with some small ships, the algorithm performance is negatively affected. To address the above problems, a framework for ship tracking in hazy weather using deep learning methods is proposed in this study. First, a lightweight convolutional neural network (CNN) is used to remove haze from images. Second, the YOLOv5 algorithm with a small target detection layer added is used to detect ships in dehazed marine images, and the Deep SORT algorithm is used to track ships. Then, multi-target tracking evaluation metrics are used to evaluate the tracking effectiveness of ships in videos taken in hazy environments and ships in videos after removal of haze. The experiments show that the PSNR of the dehazed images is 23.86, and the SSIM of the dehazed images is 0.96, indicating that the framework can effectively improve the quality of hazy maritime images. The evaluation metrics for multi-objective tracking also show that for hazy videos with small ships, the ship tracking algorithm is more robust when applied after haze removal.**

Keywords—image dehaze, ship detection, ship tracking

I. Introduction

With the rapid development of sensor technology, information acquisition, and transmission based on optical sensors are widely used in maritime intelligent transportation systems [1], such as the use of optical sensors to collect and digitally reconstruct the navigation information of maritime vessels is the basis of ship navigation situational awareness and an important prerequisite for the safe navigation of unmanned vessels [2]. As the maritime visual data collected by optical sensors has more information, accurate and stable monitoring and tracking of ship targets in maritime visual data is important to ensure the safety of ship navigation and enhance maritime monitoring, etc. However, bad weather such as haze can affect the quality of maritime images, and accurate and stable tracking of targets in maritime images in a hazy environment is still a technical challenge for intelligent ship visual navigation and control[3].

Considering the impact of haze on advanced visual tasks such as ship tracking, and further investigating the applicability of deep learning-based haze removal methods in advanced vision tasks at sea, this study proposes a deep learning-based haze removal and ship target tracking framework for maritime images and evaluates the tracking performance on a self-built simulation dataset using multi-target tracking evaluation parameters. The contributions of this study can be summarized as follows:

- This study improves YOLOv5 by adding a small target detection layer to YOLOv5 to improve the detection accuracy for small ships in the scene.
- To address the issue of the image becoming dark after haze removal, thereby obscuring the ship's target features, we improved AOD-Net [4]at the pixel level.
- This study achieves stable tracking of maritime haze scenes and provides ideas for the application of advanced vision tasks in hazy environments.

II. Related Work

A. Haze Image Enhancement in Advanced Vision Tasks

Currently, many studies are devoted to haze image enhancement in advanced vision tasks. Kun Han et al. [5] argue that the haze scenes will bring negative impacts to the object tracking process and propose an end-to-end Dehazing Siamese Region Proposal Network (DH-SiamRPN) in their study. Xin Nie et al.[6] consider that bad weather conditions can negatively affect the application of the target detection algorithm, so in their study, they use a dataset containing low-quality images to train the YOLOv3 algorithm and thus improve its detection accuracy in bad weather. To recover haze images to clear images, Qiaoling Shu et al.[7] propose to exploit the multichannel total variation (MTV) regularizer to guarantee the robust restoration of haze-free images. Jian Zhang et al.[8] propose that image processing technology is one of the key technologies for the development of IoT technology and sensible dehazing effect can even ensure the safety of navigation. In their study, a compensation textures dehazing method is presented for water alike scenes. Huimin Lu et al.[9] remove the haze caused by water turbidity from the captured images, and then use the YOLO algorithm to identify and track marine organisms. Gong Li et al. [10] conclude that the tracking method based on an unstable single feature has a poor tracking effect, so a tracking algorithm based on multi-feature fusion and spatial-context correlation is proposed to enhance the adaptability of the algorithm to environmental changes. In [11] , to achieve stable tracking of vehicles in haze scenes, a novel filtering algorithm is proposed that combines low-rank sparse and adaptive dictionary learning template with the classical particle filter algorithm. However, most of these methods have been applied to land-based scenes, and there is a need for an improved haze removal network specifically tailored for maritime haze videos, considering the differences in sea surface scattering and other imaging characteristics compared to land-based haze. Therefore, this study enhances AOD-Net for maritime haze scenes to more efficiently remove haze from maritime haze images.

B. Target Tracking

This work was jointly supported by the National Key Research and Development Program of China under Grant 2021YFC2801004 and 2021YFC2801003, National Natural Science Foundation of China under Grant 52102397, 51709167 and 52071199, China Postdoctoral Science Foundation under Grant 2021M700790. (Corresponding author: Jiansen Zhao).

979-8-3503-0853-2/23/$31.00 ©2023 IEEE

Authorized licensed use limited to: Siddhartha Academy of Higher Education Deemed To Be University. Downloaded on February 05,2026 at 09:32:17 UTC from IEEE Xplore. Restrictions apply

Target tracking algorithms as an important topic in computer vision, and there are many researches involving target tracking in maritime images. Xinqiang Chen et al.[12] propose a ship tracking framework based on nuclear correlation filter (KCF) and curve fitting, and experiments show that the framework can achieve stable tracking of ships in maritime videos. For the problems of deformation, blurring and occlusion of ship movement at sea, Huanian Liao et al. [13] propose a twin-neural network-based ship tracking method, and the experiments showed that the algorithm has high accuracy and real-time performance. To solve the problem of real-time tracking of ship movement by smart ships, a hierarchical model predictive control(H-MPC) tracking method based on the combination of model predictive control(MPC) and hierarchical control is proposed in [14]. In this paper, we adopt Deep SORT, a flexible and robust tracking model.

III. Method

A. Lightweight End-To-End CNN for Haze Removal

To improve the quality of marine images in complex scenes, marine haze images were used to train AOD-Net[4], which can achieve end-to-end dehazing to enhance the generalization performance of the network in marine scenes. At the same time, Eq. (1) was introduced to further highlight the structural features of ships in the images.

$$G(x) = mJ(x) + n \tag{1}$$

Wherein, $J(x)$ is the pixel value of the source image, $G(x)$ is the pixel value of the output image, $m > 0$ is the gain parameter, and n is the bias parameter, which is used to adjust the contrast and brightness of the marine image to further eliminate the impact of background noise. In this study, the AOD-Net model after marine scene enhancement is called e-AOD-Net.

The e-AOD-Net uses a CNN to remove haze based on the atmospheric scattering model. It can input frames of hazy videos into a CNN for hazy removal and output clear and high-quality ship images in real time. The traditional atmospheric scattering model that generates hazy images is described as follows:

$$I(x) = J(x)t(x) + A(1 - t(x)) \tag{2}$$

$$t(x) = e^{-\beta d(x)} \tag{3}$$

In the Eq. (2), $I(x)$ is the hazy image, $J(x)$ is the clear image after removing haze, $t(x)$ is the medium transmission, describing the light without scattering and transmitted to the visual sensor, which can be expressed by the atmospheric scattering coefficient β and the distance between the field and the visual sensor $d(x)$.

Eq. (2) and Eq. (3) can be transformed into:

$$J(x) = K(x)I(x) - K(x) + b \tag{4}$$

$$K(x) = \frac{\frac{1}{t(x)}(I(x) - A) + (A - b)}{I(x) - 1} \tag{5}$$

Where, b is the deviation value whose default value is 1. Meanwhile, Eq. (4) integrates $\frac{1}{t(x)}$ and A into $K(x)$.

B. Ship Tracking with Improved YOLOv5 and Deep SORT Algorithms

The YOLOv5 network can globally process the input image to effectively acquire image features [15], and has a network structure with good generalization capability, which is an ideal algorithm for implementing target detection in mobile environments. Since the ship targets in maritime images are usually small and the down-sampling times of the YOLOv5 algorithm are relatively large, it is difficult for the algorithm to learn the feature information of small targets from deeper feature maps, so this study adds a small target detection layer to the original YOLOv5 algorithm and then stitches the shallower feature maps with the deeper feature maps for detection to improve the detection accuracy of small targets in images. The network diagram is shown in Fig. 1. The dotted line in the structure diagram indicates the deleted data flow and the red line indicates the direction of the added data flow in the figure.

The video stream with the information of the detection frame is used as the input of the Deep SORT algorithm. the Deep SORT algorithm[16] first predicts the ship's trajectory in the next frame using the Kalman filter and matches the predicted information with the information of the detection frame, and successfully matches the information to track the trajectory of the target ship between the before and after frames[17].

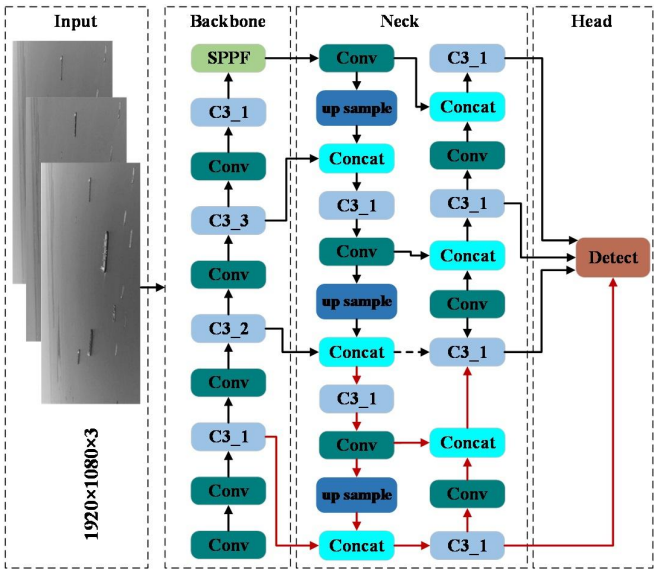

Fig. 1. YOLOv5 network structure with small target detection layer added

IV. Experiments

We validate the performance of the proposed framework for haze removal and ship tracking on a self-constructed maritime haze image dataset and also evaluate the tracking performance of the method using multi-objective tracking evaluation parameters.

A. Experimental Data

The data used in this study were taken by DJI UAV on April 5, 2021, at Baosteel Wharf, and the resolution of the image data is 1920 × 1080. To verify the performance of the

979-8-3503-0853-2/23/$31.00 ©2023 IEEE

Authorized licensed use limited to: Siddhartha Academy of Higher Education Deemed To Be University. Downloaded on February 05,2026 at 09:32:17 UTC from IEEE Xplore. Restrictions apply

proposed framework in this paper for haze removal and ship tracking, the algorithm used in [18] for synthesizing haze images is used to synthesize the haze dataset in this study. The experimental scenarios are shown in Fig. 2, and the synthetic haze dataset is shown in Figure 3.

(a) (b)

Fig. 2. experimental scenarios. (a) Scene 1 with multiple small ships. (b) Scene 2 with low illumination

(a) (b)

Fig. 3. Synthetic haze images. (a) Synthetic images in Scene 1 with multiple small ships. (b) Synthetic images in Scene 2 with low illumination

B. Evaluation Criteria

This study uses several evaluation parameters to assess the performance of the proposed framework for haze removal and ship tracking, and each parameter is defined as follows:

- PSNR: The peak signal-to-noise ratio[19] is a widely used evaluation index for measuring image quality.
- SSIM: The structural similarity index measure [20] can objectively determine the structural similarity of images based on the human visual system.
- IDF1[21]: The ratio of correctly identified detections over the average number of ground-truth and computed detections.
- MOTA (multi-object tracking accuracy) [22] : This measure combines three error sources: false positives, missed targets and identity switches.
- MOTP(Multiple Object Tracking Precision) [22] : The misalignment between the annotated and the predicted bounding boxes.
- MT: The ratio of ground-truth trajectories that are covered by a track hypothesis for at least 80% of their respective life span.
- ML: The ratio of ground-truth trajectories that are covered by at rack hypothesis for at most 20% of their respective life span.
- FP: The total number of false positives; FN: The total number of false negatives (missed targets).
- FN: The total number of false negatives (missed targets).
- IDS: The total number of identity switches.

C. Experimental Process and Results

1) Haze removal process and results of maritime images: To enhance the clarity of maritime haze images and to eliminate haze noise to highlight the structural features of ships in the images, e-AOD-Net was used in this study to remove haze from maritime images, and evaluation parameters were used to assess the image quality after haze removal. The results of haze removal are shown in Fig. 4 and Fig. 5, and the evaluation table is shown in Table 1.

(a) (b) (c)

Fig. 4. The results of images haze removal in scenario 1

(a) (b) (c)

Fig. 5. The results of images haze removal in scenario 2

TABLE I. EVALUATION RESULTS OF DEHAZED IMAGES

Method	e-AOD- Net	
	PSNR	***SSIM***
scenario 1	23.84	0.96
scenario 2	23.71	0.95

It can be seen that images after removing haze is closer to the original image, and the features of the ships in images are more prominent, which is more conducive to the target tracking algorithm for learning and detecting the features of the target in images.

2) Ship tracking process and results: We adopted the YOLOv5 algorithm with a small target detection layer added to detect ships in the synthesized hazy images and images after haze removal. During the maritime target tracking experiment, 4000 self-built image sets were divided into 3200 training datasets, 400 validation datasets, and 400 test datasets to train the YOLOv5 algorithm with a small target detection layer added. YOLOv5 is used as a comparison method to verify the detection performance of the improved YOLOv5 algorithm with a small target detection layer added in this paper. The training comparison chart is shown in Fig. 6 and Fig. 7, the train metrics, validation metrics and test metrics is shown in Table 2, 3 and 4, and the detecting results are shown in Fig. 8 and Fig. 9.

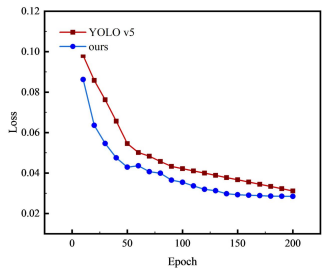

Fig. 6. Train carve

979-8-3503-0853-2/23/$31.00 ©2023 IEEE

Authorized licensed use limited to: Siddhartha Academy of Higher Education Deemed To Be University. Downloaded on February 05,2026 at 09:32:17 UTC from IEEE Xplore. Restrictions apply

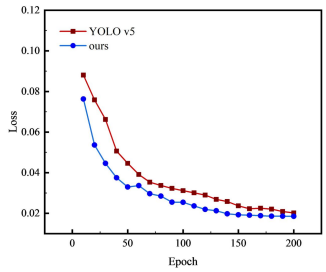

Fig. 7. Validation carve

TABLE II. TRAIN METRICS OF YOLOV5 AND YOLOV5 WITH SMALL TARGET DETECTION LAYER ADDED TRAINING PROCESS

	P	R	mAP_0.5	mAP_0.5:0.95
YOLOv5	0.981	0.9701	0.982	0.9103
OURS	0.9899	0.972	0.988	0. 918

TABLE III. VALIDATION METRICS OF YOLOV5 AND YOLOV5 WITH SMALL TARGET DETECTION LAYER ADDED TRAINING PROCESS

	P	R	mAP_0.5	mAP_0.5:0.95
YOLOv5	0.973	0.982	0.995	0.845
OURS	0.985	0.989	0.991	0.879

TABLE IV. TEST METRICS OF YOLOV5 AND YOLOV5 WITH A SMALL TARGET DETECTION LAYER ADDED TRAINING PROCESS

	P	R	mAP_0.5	mAP_0.5:0.95
YOLOv5	0.986	0.979	0.989	0.81
OURS	0.990	0.980	0.991	0.821

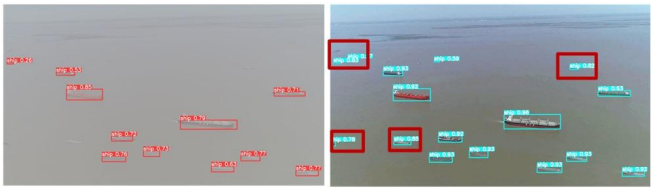

Fig. 8. Comparison of detection results of images in scenario 1 before and after haze removal using the YOLOv5 algorithm with a small target detection layer added. (Detection details are highlighted with red boxes).

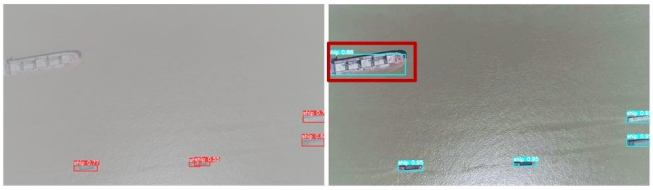

Fig. 9. Comparison of detection results of images in scenario 2 before and after haze removal using the YOLOv5 algorithm with a small target detection layer added. (Detection details are highlighted with red boxes).

As can be seen from Tables 2, 3 and 4, the performance differences between the improved YOLOv5 and YOLOv5 are minimal, but the YOLOv5 with a small target detection layer added in this paper has improved in terms of detection accuracy and recall. Fig. 8 and Fig. 9 show that YOLOv5 with a small target detection layer added has high-precision detection performance in both two scenarios. However, the degree of recovery of the images after haze removal is high, and the structural features of the ships are prominent, making it easy for the ship target in the image to be detected by the target detection algorithm, such as the small ships in the red boxes in scenario 1. Haze noise in images reduces the accuracy of the target detection. In the case of high haze concentrations, some ships were not detected, such as those in the red boxes in Scene 2.

In this study, the Deep SORT algorithm was adopted as the target tracking algorithm to track ships in maritime images to achieve high accuracy ship tracking in a haze environment, and the SORT algorithm as a comparison method to verify the tracking performance of the algorithm. In this study, the tracking performance of the algorithm was evaluated in the images before and after haze removal, the tracking evaluation results are shown in Table 4 and Table 5.

TABLE V. EVALUATION RESULTS OF MULTI-SHIP TRACKING IN SCENARIO 1

scenario 1									
Data type	***Method***	***IDF1***	***MOTA***	***MOTP***	***MT***	***ML***	***FP***	***FN***	***IDS***
Hazy video	SORT	89.3%	88.7%	84.2%	75.0 %	12.5 %	24	292	3
	Deep SORT	96.8%	93.8%	84.7%	75.0 %	12.5 %	23	155	3
Dehazy video	SORT	97.7%	96.8%	86.0%	100. 0%	0.0%	13	70	1
	Deep SORT	**98.7%**	**97.5%**	**85.7%**	**100. 0%**	**0.0%**	**3**	**44**	**0**

TABLE VI. EVALUATION RESULTS OF MULTI-SHIP TRACKING IN SCENARIO 2

scenario 2									
Data type	***Method***	***IDF1***	***MOTA***	***MOTP***	***MT***	***ML***	***FP***	***FN***	***IDS***
Hazy video	SORT	92.7%	85.0%	80.4%	87.5 %	0.0%	82	84	2
	Deep SORT	97.6%	96.7%	85.4%	100. 0%	0.0%	18	72	1
Dehazy video	SORT	97.6%	93.2%	81.8%	87.5 %	0.0%	54	149	0
	Deep SORT	**98.7%**	**97.5%**	**85.8%**	**100. 0%**	**0.0%**	**3**	**44**	**0**

As shown in Tables 4 and 5, the Deep SORT algorithm combined with the tracking algorithm of YOLOv5 with a small target detection layer added have high MOTA and MOTP values in the tracking evaluation and its IDS and ML values are low. This shows that the algorithm in this study can track small ships stably between consecutive frames while avoiding the number of ID transitions. Besides, the evaluation results of tracking ships in maritime images after haze removal are usually better values, indicating that the method of haze removal used in this study can effectively improve the quality of the images and at the same time can effectively improve the robustness of the target tracking algorithm in maritime scenes.

V. CONCLUSION

To validate the application of deep learning-based methods in improving the detection and tracking accuracy of small vessels in hazy environments, a small target detection layer is added to the YOLOv5 algorithm in this study, and a lightweight CNN is used to enhance the quality of maritime images in hazy environment scenes, and then the ships in maritime videos in the hazy environment are tracked, and multi-objective evaluation parameters are employed to evaluate the performance of the tracking algorithm. The experiments show that the PSNR of the dehazed images is 23.86, and the SSIM of the dehazed images is 0.96, indicating that the framework can effectively improve the quality of hazy maritime images. The evaluation metrics of multi-objective tracking also show the higher robustness of

979-8-3503-0853-2/23/$31.00 ©2023 IEEE

Authorized licensed use limited to: Siddhartha Academy of Higher Education Deemed To Be University. Downloaded on February 05,2026 at 09:32:17 UTC from IEEE Xplore. Restrictions apply

the ship tracking algorithm after the haze removal process for videos in hazy environments with some small ships.

REFERENCES

[1] Y. Lu, Y. Guo, and M. Liang, "CNN-Enabled Visibility Enhancement Framework for Vessel Detection under Haze Environment," J Adv Transp, vol. 2021, 2021, doi: 10.1155/2021/5598390.

[2] R. W. Liu, Y. Guo, Y. Lu, K. T. Chui, and B. B. Gupta, "Deep Network-Enabled Haze Visibility Enhancement for Visual IoT-Driven Intelligent Transportation Systems," IEEE Trans Industr Inform, vol. 19, no. 2, pp. 1581–1591, Feb. 2023, doi: 10.1109/TII.2022.3170594.

[3] A. Artusi and K. A. Raftopoulos, "A Framework for Objective Evaluation of Single Image De-Hazing Techniques," IEEE Access, vol. 9, pp. 76564–76575, 2021, doi: 10.1109/ACCESS.2021.3082207.

[4] B. Li, X. Peng, Z. Wang, J. Xu, and D. Feng, "AOD-Net: All-in-One Dehazing Network," Proceedings of the IEEE International Conference on Computer Vision, vol. 2017-Octob, pp. 4780–4788, 2017, doi: 10.1109/ICCV.2017.511.

[5] K. Han, J. Peng, Q. Yang, and W. Tian, "An End-to-End Dehazing Siamese Region Proposal Network for High Robustness Object Tracking," IEEE Access, vol. 9, pp. 91983–91994, 2021, doi: 10.1109/ACCESS.2021.3091434.

[6] X. Nie, M. Yang, and R. W. Liu, "Deep Neural Network-Based Robust Ship Detection Under Different Weather Conditions," in 2019 IEEE Intelligent Transportation Systems Conference (ITSC), 2019, pp. 47–52. doi: 10.1109/ITSC.2019.8917475.

[7] Q. Shu, C. Wu, Q. Zhong, and R. W. Liu, "Alternating minimization algorithm for hybrid regularized variational image dehazing," Optik (Stuttg), vol. 185, pp. 943–956, May 2019, doi: 10.1016/j.ijleo.2019.04.002.

[8] J. Zhang, F. Feng, and W. Song, "A compensation textures dehazing method for water alike area," Journal of Supercomputing, vol. 77, no. 4, pp. 3555–3570, Apr. 2021, doi: 10.1007/s11227-020-03406-8.

[9] H. Lu, T. Uemura, D. Wang, J. Zhu, Z. Huang, and H. Kim, "Deep-Sea Organisms Tracking Using Dehazing and Deep Learning," Mobile Networks and Applications, vol. 25, no. 3, pp. 1008–1015, Jun. 2020, doi: 10.1007/s11036-018-1117-9.

[10] G. Li, Z. yang Wang, J. Luo, X. Chen, and H. biao Li, "Spatio-Context-Based Target Tracking with Adaptive Multi-Feature Fusion for Real-World Hazy Scenes," Cognit Comput, vol. 10, no. 4, pp. 545–557, Aug. 2018, doi: 10.1007/s12559-018-9550-4.

[11] W. Gang, Z. Xiaoqin, S. Shoubao, and L. Hong, "Vehicle Tracking Incorporating Low-Rank Sparse into Particle Filter in Haze Scene," in Proceedings - 2016 3rd International Conference on Information Science and Control Engineering, ICISCE 2016, Institute of Electrical and Electronics Engineers Inc., Oct. 2016, pp. 739–742. doi: 10.1109/ICISCE.2016.163.

X. Chen, X. Xu, Y. Yang, O. Postolache, Z. Yu, and L. Qi, "Accurate Ship Tracking in Maritime Images with Kernelized Correlation Filter and Curve Fitting," in 2019 International Conference on Sensing and Instrumentation in IoT Era (ISSI), 2019, pp. 1–6. doi: 10.1109/ISSI47111.2019.9043651.

H. Liao, W. Wang, S. Wang, and X. Xu, "Multi-Scale Ship Tracking Based on Maritime Monitoring Platform," in Proceedings of the 15th IEEE Conference on Industrial Electronics and Applications, ICIEA 2020, Institute of Electrical and Electronics Engineers Inc., Nov. 2020, pp. 945–949. doi: 10.1109/ICIEA48937.2020.9248235.

Y. Teng, Z. Liu, L. Zhang, C. Liu, and N. Wei, "On Intelligent Ship Tracking Control Method Based on Hierarchical MPC," in Chinese Control Conference, CCC, IEEE Computer Society, Jul. 2021, pp. 2814–2819. doi: 10.23919/CCC52363.2021.9550295.

J. H. Kim, N. Kim, Y. W. Park, and C. S. Won, "Object Detection and Classification Based on YOLO-V5 with Improved Maritime Dataset," J Mar Sci Eng, vol. 10, no. 3, 2022, doi: 10.3390/jmse10030377.

N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," Proceedings - International Conference on Image Processing, ICIP, vol. 2017-Septe. pp. 3645–3649, 2018. doi: 10.1109/ICIP.2017.8296962.

J. Zhao, Y. Chen, Z. Zhou, J. Zhao, S. Wang, and X. Chen, "Extracting Vessel Speed Based on Machine Learning and Drone Images during Ship Traffic Flow Prediction," J Adv Transp, vol. 2022, 2022, doi: 10.1155/2022/3048611.

R. W. Liu, W. Yuan, X. Chen, and Y. Lu, "An enhanced CNN-enabled learning method for promoting ship detection in maritime surveillance system," Ocean Engineering, vol. 235, no. June, p. 109435, 2021, doi: 10.1016/j.oceaneng.2021.109435.

Z. Wang and A. C. Bovik, "Error : Love It or Leave It ?," no. January, pp. 98–117, 2009.

Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," IEEE Transactions on Image Processing, vol. 13, no. 4, pp. 600–612, 2004, doi: 10.1109/TIP.2003.819861.

E. Ristani, F. Solera, R. S. Zou, R. Cucchiara, and C. Tomasi, "Performance Measures and a Data Set for Multi-Target, Multi-Camera Tracking," Sep. 2016, [Online]. Available: http://arxiv.org/abs/1609.01775

K. Bernardin and R. Stiefelhagen, "Evaluating multiple object tracking performance: The CLEAR MOT metrics," EURASIP J Image Video Process, vol. 2008, 2008, doi: 10.1155/2008/246309.

979-8-3503-0853-2/23/$31.00 ©2023 IEEE

Authorized licensed use limited to: Siddhartha Academy of Higher Education Deemed To Be University. Downloaded on February 05,2026 at 09:32:17 UTC from IEEE Xplore. Restrictions apply